JAMHURIYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY (CIT)

LINUX COMMANDS ASSIGNMENT

Chapters 1 – 4

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Course: Linux

Assignment Topic: Linux Commands and Practical Exercises

Submitted To: [Nasri]

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CHAPTER 1: INTRODUCTION TO LINUX AND UBUNTU

1.1 Introduction

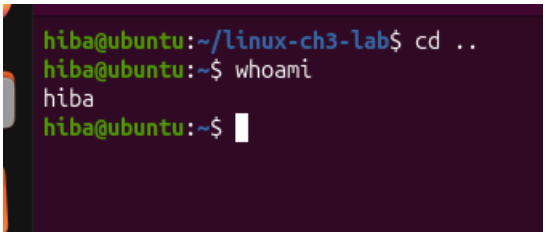
This chapter introduces Linux and Ubuntu. It explains the Linux operating system and provides practical commands used in the Ubuntu terminal.

1.2 Linux Commands and Their Explanations

Command 1: whoami

Explanation: Displays the name of the user currently running the shell.

Example: To display the username of the currently logged-in user, type whoami in the terminal.

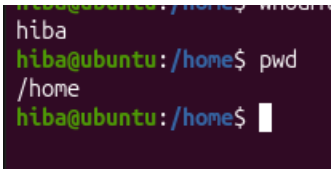


Expected result: The terminal displays the current username.

Command 2: pwd

Explanation: Displays the current working directory as an absolute path.

Example: To display the current working directory (the location where you are currently working), type pwd in the terminal.

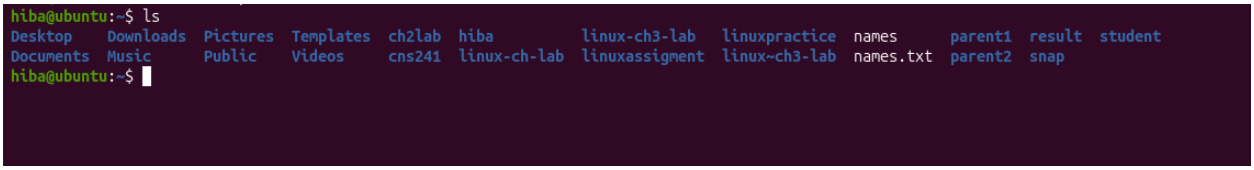


Expected result: The terminal displays the files and directories in the current directory.

Command 3: ls

Explanation: Displays the contents of the current directory.

Example: To list the files and directories in the current working directory, type ls in the terminal.



Expected result: The terminal displays the files and directories in the current directory.

Command 4: cat /etc/os-releas

Explanation: Displays the contents of the /etc/os-release file.

Example: To display information about the installed Linux operating system, such as its name and version, type cat /etc/os-release in the terminal.

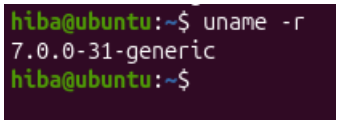


Expected result: The terminal displays information about the installed Linux operating system.

Command 5: uname -r

Explanation: Displays the release version of the running Linux kernel.

Example: To display the version of the Linux kernel currently running on the system, type uname -r in the terminal.



Expected result: The terminal displays the running Linux kernel version.

CHAPTER 2: LINUX SYSTEM ARCHITECTURE, COMMAND LINE AND FILE MANAGEMENT

2.1 Introduction

This chapter introduces Linux system architecture, the command line interface, Bash shell, and file management. It explains how to navigate the Linux filesystem, obtain help, use command syntax, and create, copy, move, and remove files and directories using the Ubuntu termina

2.2 Linux Commands and Their Explanations

Command 1: echo $SHELL

Explanation: Displays the name and path of the shell currently being used.

Example: To display the name and path of the current shell being used, type echo $SHELL in the terminal.

```
hiba@ubuntu:~$ echo $SHELL
/bin/bash
hiba@ubuntu:~$
```

Expected Result: The terminal displays /bin/bash.

Command 2: history

Explanation: Displays a list of previously executed commands.

Example: To display a list of previously executed commands in the terminal, type history in the terminal.

```
106  cat names.txt
107  nano names.txt
108  cd ~
109  mkdir -p ~/linux~ch3-lab
110  cd ~/linux-ch3-lab
111  printf ,Ali/nHodan/nYusuf/n' > names. txt
112  cat names.txt
113  echo Hodan > names.txt
114  echo ali names.txt
115  echo yusuf names.txt
116  cat names.txt
117  clar
118  cd
119  cd~
120  cd ~
121  cd ..
122  whoami
123  cd ..
124  whoami
125  pwd
126  cd ..
127  pwd
128  ls
129  mkdir practice
130  touch names.txt
131  echo "my name is hiba" > names.txt
132  pwd
133  mkdir names
134  cd ..
135  pwd
136  mkdir names
137  qcd ~
138  cd ~
139  pwd
140  mkdir names
141  touch names.txt
142  echo "my name is hiba"> names.txt
143  cat names.txt
144  echo $shell
145  echo $SHELL
146  history
ba@ubuntu:~$
```

Expected Result: The terminal displays a list of previously executed commands with their command numbers.

Command 3: ls -a

Explanation: Displays all files and directories, including hidden files.

Example: To list all files and directories, including hidden files and directories, in the current working directory, type ls -a in the terminal.

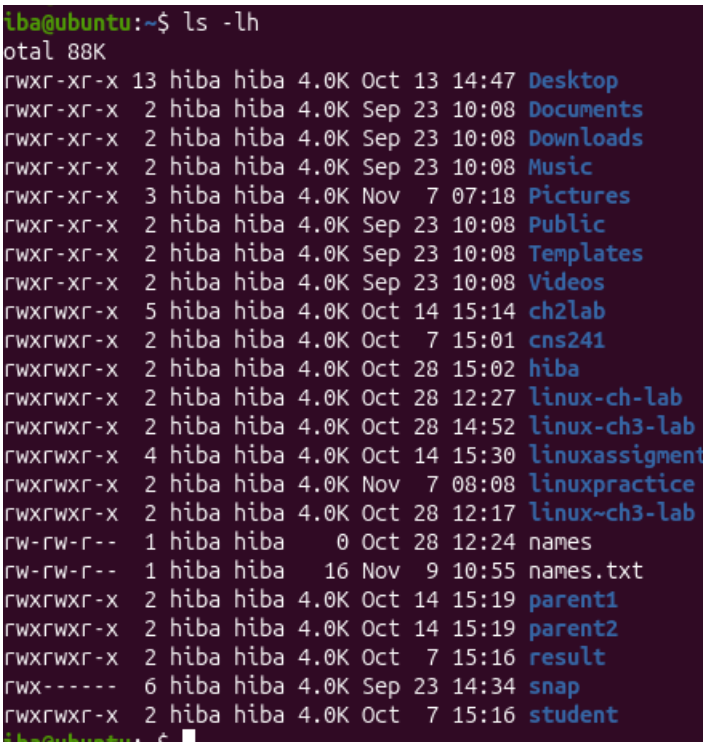


Expected Result: Displays all files and directories, including hidden files (those starting with a dot .).

Command 4: ls -lh

Explanation: Displays a detailed list with file sizes in a human-readable format.

Example: To display files and directories with detailed information, including file sizes in a human-readable format, type ls -lh in the terminal.

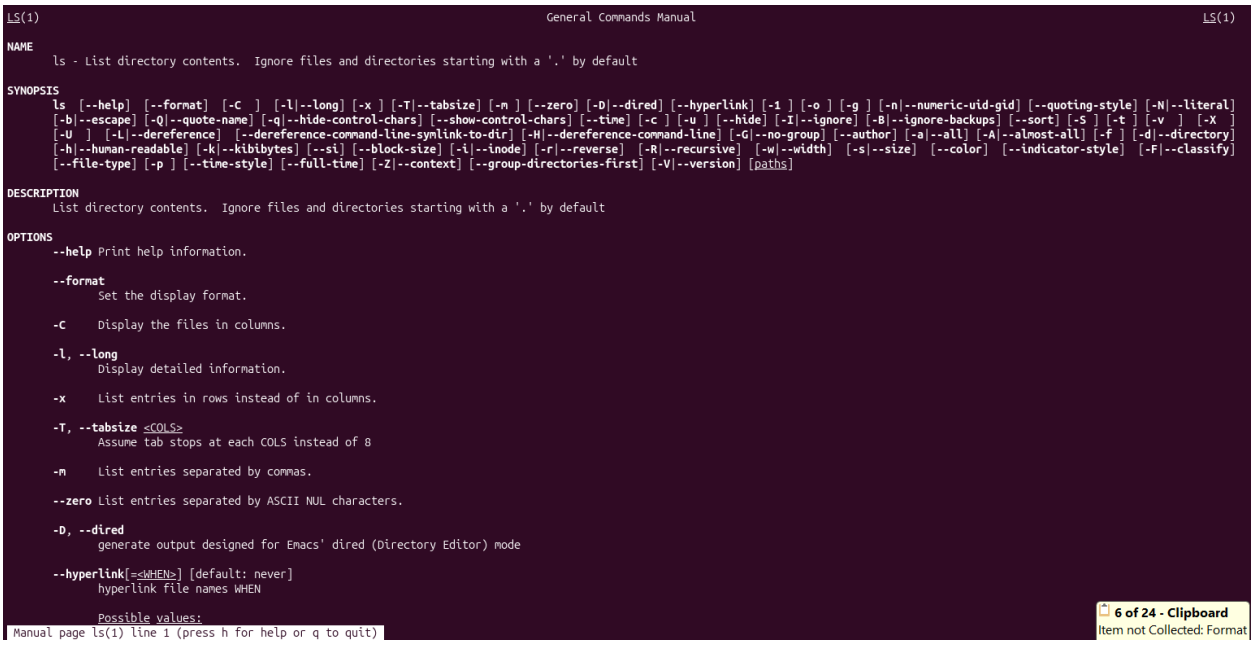


Expected Result: Displays files and directories in a detailed, human-readable format, including file sizes, permissions, and modification dates.

Command 5: man ls

Explanation: Opens the manual page for the ls command.

Example: To view the manual page and detailed documentation about the ls command, type man ls in the terminal.

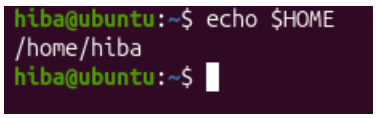


Expected Result: Displays the manual page for the ls command, showing its description, usage, options, and examples.

Command 6: echo $HOME

Explanation: Displays the home directory of the current user.

Example: To display the path of the current user's home directory, type echo $HOME in the terminal.

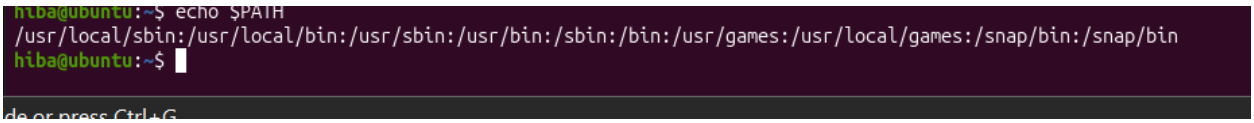


Expected Result: Displays the path of the current user's home directory, usually /home/username.

Command 7: echo $PATH

Explanation: Displays the directories where the shell searches for executable programs.

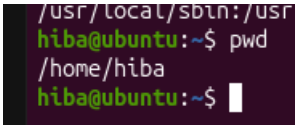
Example: To display the list of directories where the system searches for executable commands, type echo $PATH in the terminal.



Expected Result: Displays a list of directories where Linux searches for executable commands.

Command 8: pwd

Explanation: Displays the current working directory as an absolute path.

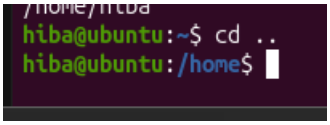


Expected Result: Displays the full path of the current working directory.

Command 9: cd ..

Explanation: Moves to the parent directory of the current directory.

Example: To move one level up to the parent directory, type cd .. in the terminal.

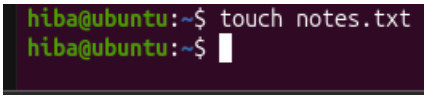


Expected Result: Moves one directory level up to the parent directory.

Command 10: touch notes.txt

Example: To create an empty file named notes.txt, type touch notes.txt in the terminal.

Explanation: Creates an empty file named notes.txt, or updates its timestamps if it already exists.



Expected Result: Creates an empty file named notes.txt in the current directory.

Command 11: mkdir week3

Example: To create a new directory named week3, type mkdir week3 in the terminal.

Explanation: Creates a directory named week3.

```
hiba@ubuntu:~$ mkdir week3
hiba@ubuntu:~$
```

Expected Result: Creates a new directory named week3 in the current working directory.

Command 12: cp notes.txt notes.bak

Explanation: Copies notes.txt to a new file named notes.bak

Example: To copy the file notes.txt and create a backup named notes.bak, type cp notes.txt notes.bak in the terminal.

```
cp: target 'bak': No such file or directory
hiba@ubuntu:~$
```

Expected Result: Copies the contents of notes.txt into a new file named notes.bak.

Command 13: mv notes.txt old.txt

Explanation: Renames notes.txt to old.txt

Example: To rename the file notes.txt to old.txt, type mv notes.txt old.txt in the terminal.

```
hiba@ubuntu:~$ mv notes.txt old.txt
hiba@ubuntu:~$
```

Expected Result: Renames the file notes.txt to old.txt in the current directory.

Command 14: rm notes.bak

Explanation: Removes the file named notes.bak.

Example: To delete the file named notes.bak, type rm notes.bak in the terminal.

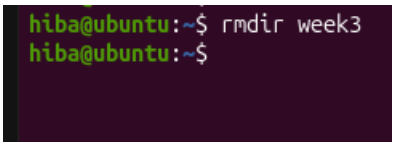
```
hiba@ubuntu:~$ rm notes.bak
hiba@ubuntu:~$
```

Expected result: The command completes without displaying output.

Command 15: rmdir week3

Explanation: Removes the directory named week3 if it is empty.

Example: To remove an empty directory named week3, type rmdir week3 in the terminal.



Expected result: The command completes without displaying output.

CHAPTER 3: TEXT PROCESSING, SEARCHING, ARCHIVING, AND EDITORS

Linux Commands and Their Explanations

3.1 Introduction

This chapter introduces Linux commands used to view and search files, process text, create archives, compress files, work with terminal text editors, and manage file links.

3.2 Linux Commands and Their Explanations

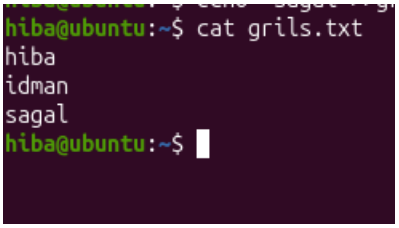
The following are the 26 command entries listed in Chapter 3.

3.2.1 Viewing and Searching Files

Command 1: cat

Explanation: Displays the contents of one or more text files. It can also combine files.

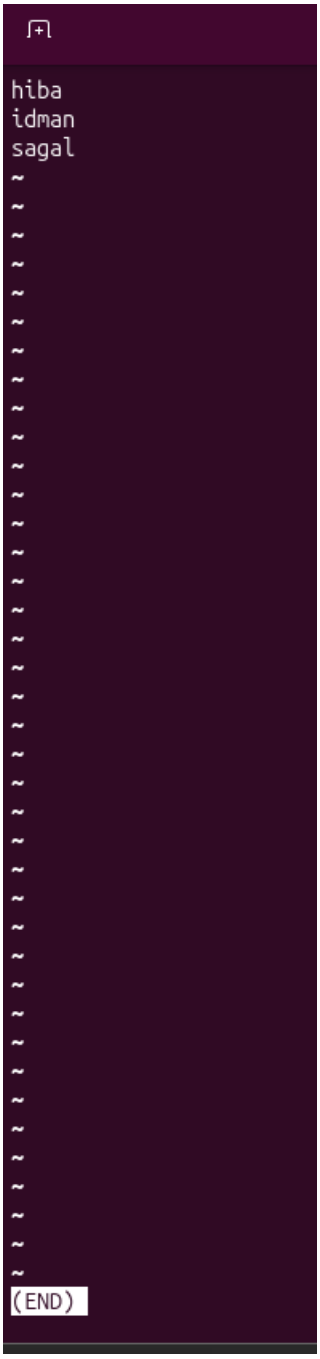
Example: `cat names.txt`



Command 2: less

Explanation: Displays a text file one screen at a time. Press q to quit.

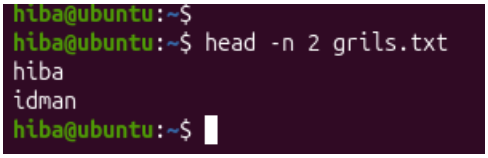
Example: `less names.txt`



Command 3: head

Explanation: Displays the first lines of a file. By default, it displays the first 10 lines.

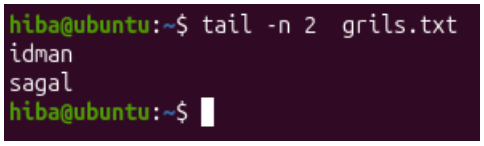
Example: `head -n 2 names.txt`



Command 4: tail

Explanation: Displays the last lines of a file. By default, it displays the last 10 lines.

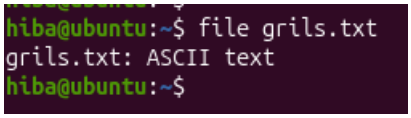
Example: `tail -n 2 names.txt`



Command 5: file

Explanation: Identifies the type of a file by examining its contents.

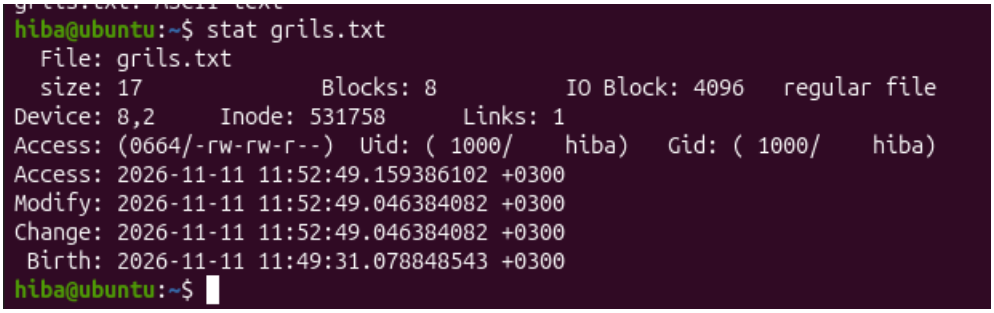
Example: `file names.txt`



Command 6: stat

Explanation: Displays detailed information about a file, such as its size, permissions, and modification time.

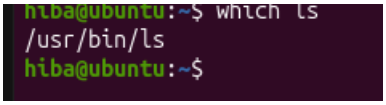
Example: `stat names.txt`



Command 7: which

Explanation: Shows the location of an executable command found in the PATH directories.

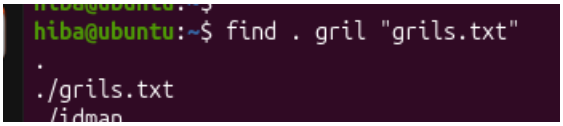
Example: `which ls`



Command 8: find

Explanation: Searches for files and directories using conditions such as name, type, and size.

Example: `find . -gril 'grils.txt'`



Command 9: locate

Explanation: Finds files quickly by searching a previously created database of file paths.

Example: `locate grils.txt`



Command 10: grep

Explanation: Searches for a text pattern inside files and displays matching lines.

Example: `grep 'Ali' names.txt`



3.2.2 Text Processing

Command 11: sort

Explanation: Sorts lines of text alphabetically or numerically.

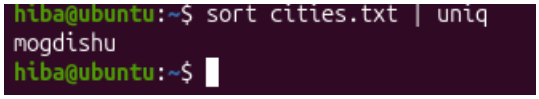
Example: `sort scores.txt`



Command 12: uniq

Explanation: Removes or reports adjacent repeated lines, commonly after sorting.

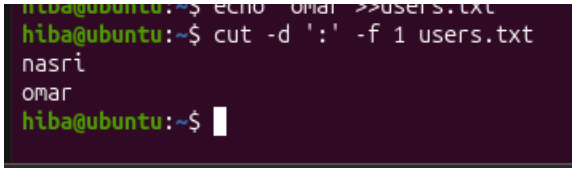
Example: `sort cities.txt | uniq`



Command 13: cut

Explanation: Extracts selected sections or fields from each line of a file.

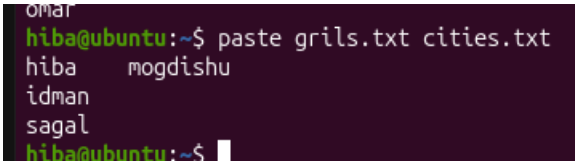
Example: `cut -d ':' -f 1 users.txt`



Command 14: paste

Explanation: Combines lines from files side by side.

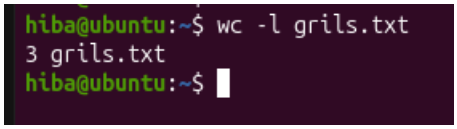
Example: `paste grils.txt cities.txt`



Command 15: wc

Explanation: Counts lines, words, and bytes or characters in a file.

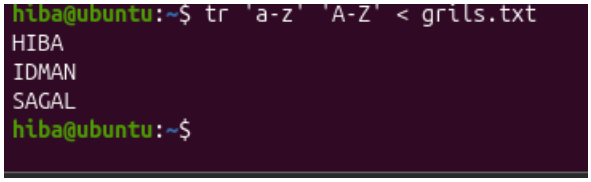
Example: `wc -l grils.txt`



Command 16: tr

Explanation: Translates or deletes characters from standard input.

Example: `tr 'a-z' 'A-Z' < grils.txt`

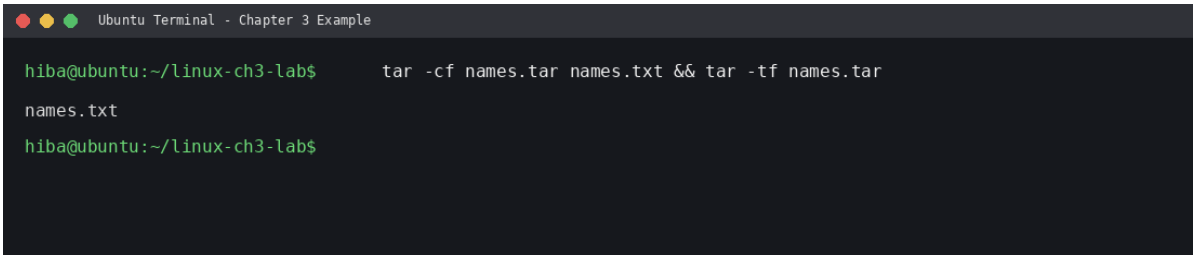


3.2.3 Archives and Compression

Command 17: tar

Explanation: Creates and extracts archive files, often used to group files together.

Example: `tar -cf names.tar names.txt && tar -tf names.tar`



Command 18: gzip

Explanation: Compresses files using gzip format. The default behavior replaces the original file with a compressed file.

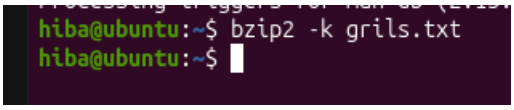
Example: `gzip -k grils.txt`



Command 19: bzip2

Explanation: Compresses files using the bzip2 compression format.

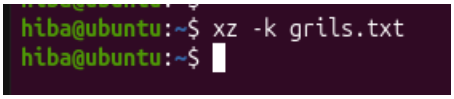
Example: `bzip2 -k names.txt`



Command 20: xz

Explanation: Compresses files using the XZ compression format.

Example: `xz -k grils.txt`



Command 21: zip

Explanation: Creates a ZIP archive containing one or more files.

Example: `zip names.zip names.txt`

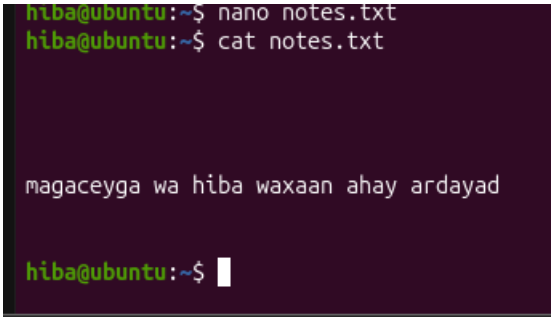


3.2.4 Text Editors: Nano and Vim

Command 22: nano

Explanation: Opens a simple terminal text editor. Use Ctrl+O to save and Ctrl+X to exit.

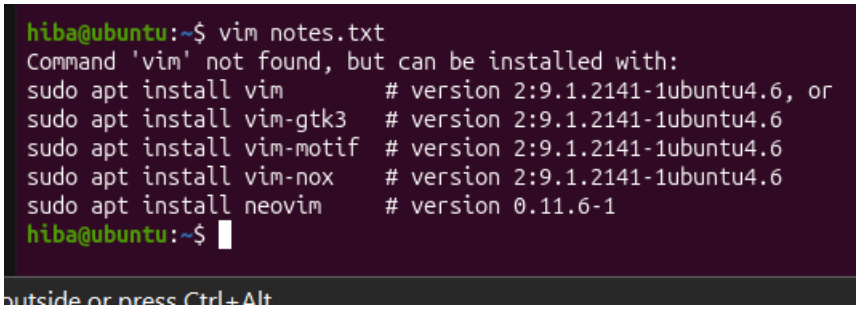
Example: `nano notes.txt`



Command 23: vim

Explanation: Opens the Vim text editor. Press i to insert text, Esc to leave insert mode, and :wq to save and quit.

Example: `vim notes.txt`

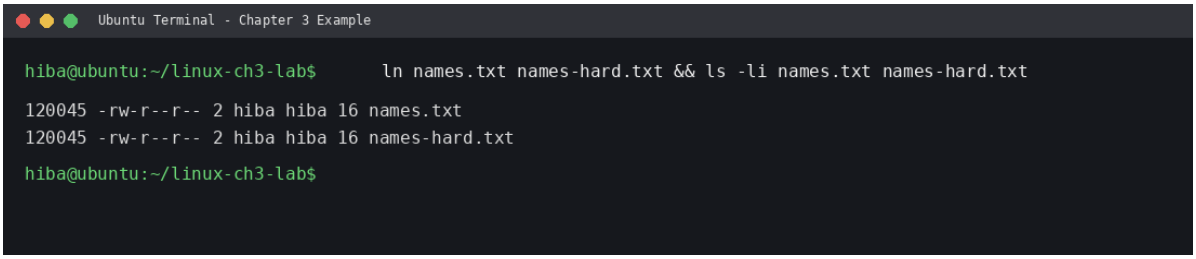


3.2.5 Hard Links and Symbolic Links

Command 24: ln

Explanation: Creates a hard link to an existing file when used without the -s option.

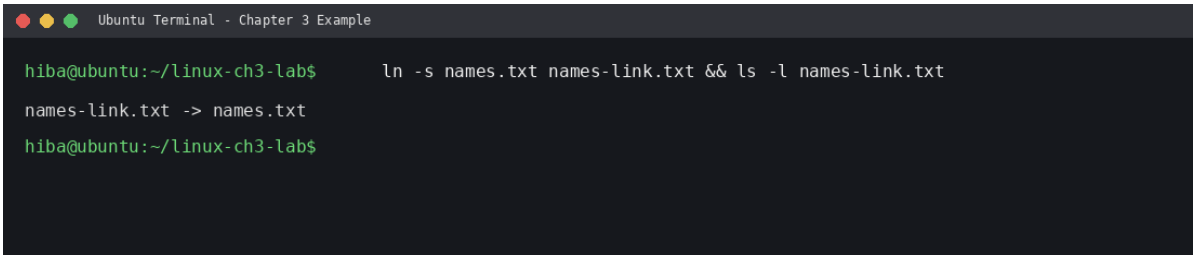
Example: `ln names.txt names-hard.txt && ls -li names.txt names-hard.txt`



Command 25: ln -s

Explanation: Creates a symbolic (soft) link that points to another file or directory.

Example: `ln -s names.txt names-link.txt && ls -l names-link.txt`



Command 26: ls -li

Explanation: Lists files with their inode numbers and detailed information.

Example: `ls -li names.txt`



CHAPTER 4: PROCESS, SERVICE AND SYSTEM MANAGEMENT

Linux / Ubuntu Commands • Explanation • Examples • Terminal illustrations

Note: The terminal panels are illustrative examples of possible output. Your output may differ depending on your Ubuntu version, user, running processes and installed services.

How to use: Copy the command shown under each topic and run it in Ubuntu Terminal. Commands that stop processes or shut down the computer should be used carefully.

4.1 Processes and Jobs

1. ps

Explanation: Displays processes currently running in the terminal.

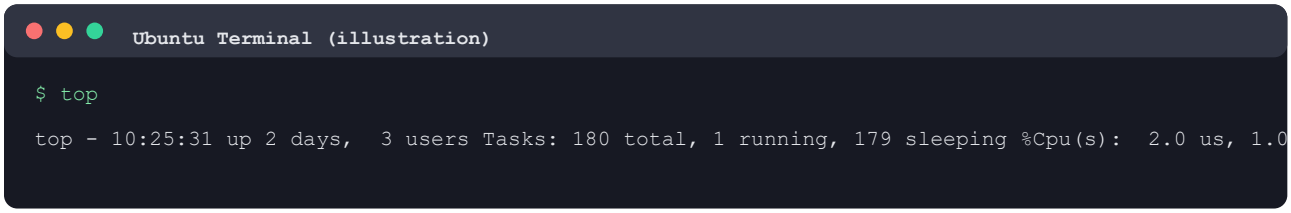
Example: `ps`

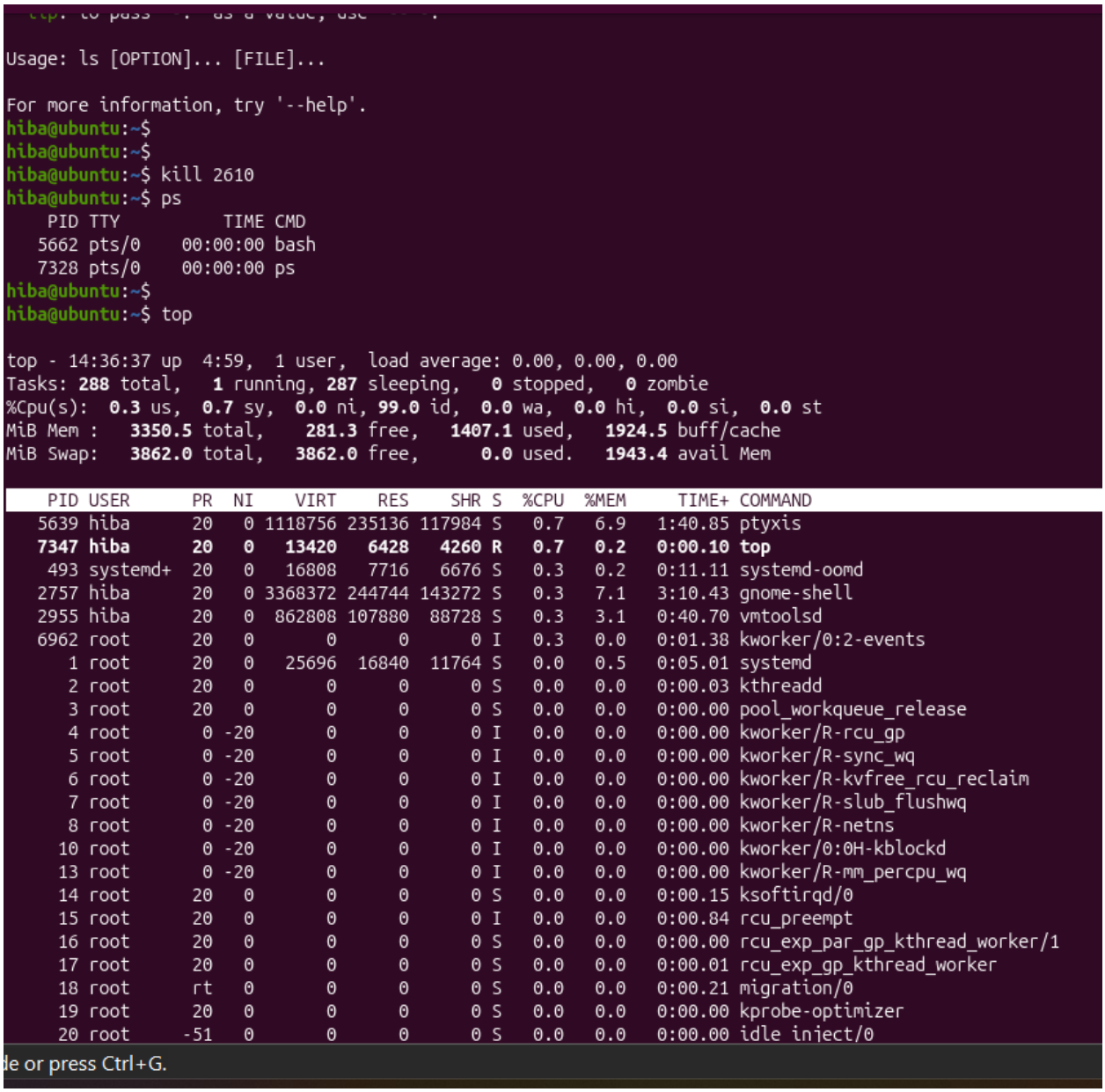
```
hiba@ubuntu:~$ ps
    PID TTY          TIME CMD
   5662 pts/0    00:00:00 bash
   7328 pts/0    00:00:00 ps
hiba@ubuntu:~$
```

2. top

Explanation: Displays running processes and system resource usage in real time.

Example: `top`

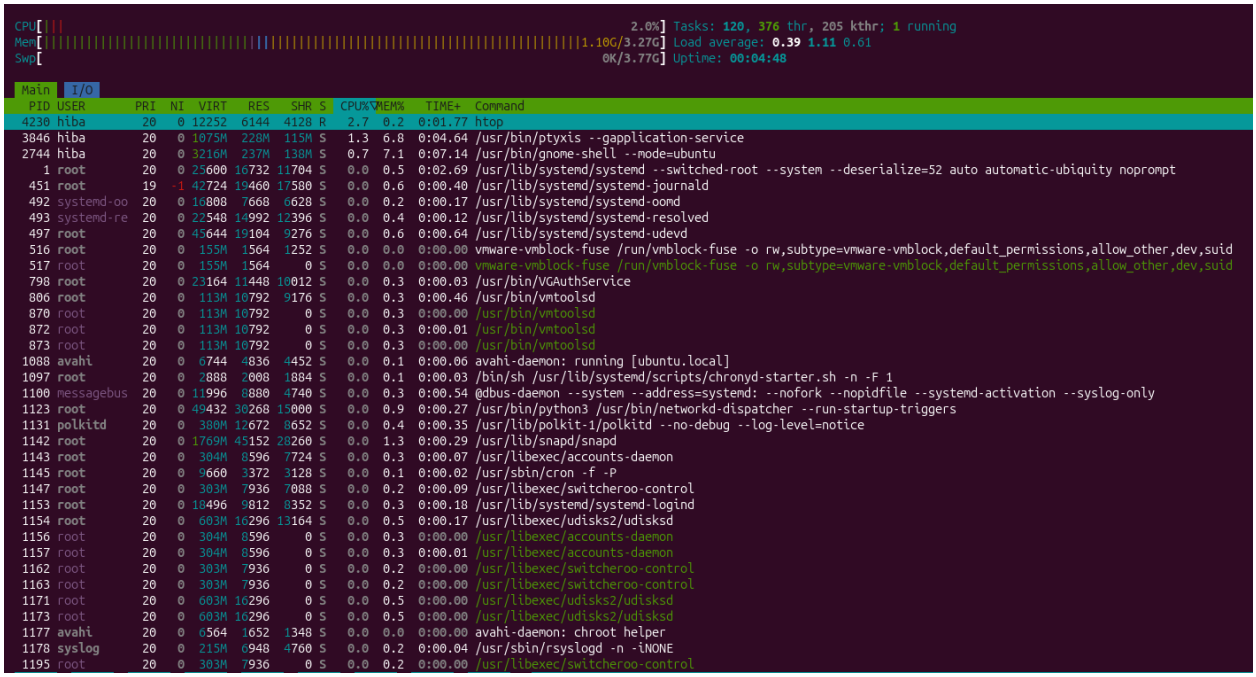




3. htop

Explanation: Displays processes in an interactive, user-friendly interface. It may need installation first.

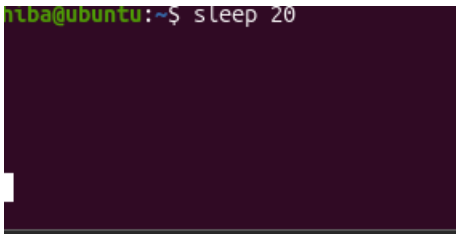
Example: `htop`



4. jobs

Explanation: Displays jobs started from the current shell, including background or stopped jobs.

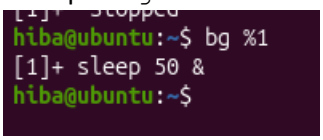
Example: `sleep 100 &; jobs`



5. bg

Explanation: Resumes a stopped job in the background.

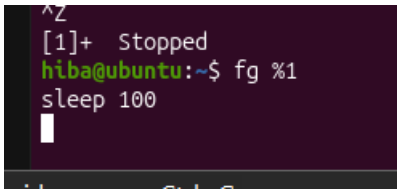
Example: `bg %1`



6. fg

Explanation: Brings a background job to the foreground.

Example: `fg %1`

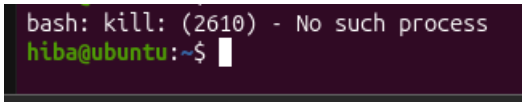


4.2 Signals, Kill and Priority

7. kill

Explanation: Sends a signal to a process identified by its process ID (PID).

Example: `kill 2610`



8. killall

Explanation: Sends a signal to processes matching a program name.

Example: `killall sleep`



9. pkill

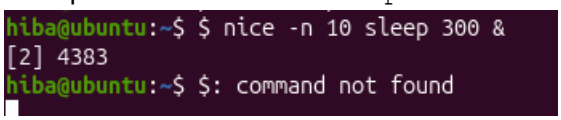
Explanation: Sends a signal to processes selected by name or other criteria.

Example: `pkill sleep`

10. nice

Explanation: Starts a program with a chosen scheduling niceness value. Higher niceness generally means lower priority.

Example: `nice -n 10 sleep 300 &`



11. renice

Explanation: Changes the niceness value of a process that is already running.

Example: `sudo renice -n 5 -p 4383`

```
renice: failed to get priority for 3301 (process ID): No such proces
hiba@ubuntu:~$ sudo renice -n 5 -p 4383
renice: failed to get priority for 4383 (process ID): No such proces
hiba@ubuntu:~$
```

e or press Ctrl+G.

4.3 System Information

12. uname

Explanation: Displays information about the operating system and kernel.

Example: `uname -a`

```
hiba@ubuntu:~$ uname -a
Linux ubuntu 7.0.0-31-generic #31-Ubuntu SMP PREEMPT_DYNAMIC Sat Aug  1 04:26:38 UTC 2026 x86_64 GNU/Linux
hiba@ubuntu:~$
```

13. uptime

Explanation: Shows how long the system has been running, plus users and load averages.

Example: `uptime`

```
hiba@ubuntu:~$ uptime
 20:33:22 up 22 min,  1 user,  load average: 0.01, 0.17, 0.28
hiba@ubuntu:~$
```

14. free

Explanation: Displays RAM and swap memory usage.

Example: `free -h`

```
hiba@ubuntu:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.3Gi       1.3Gi       256Mi        19Mi       1.9Gi       1.9Gi
Swap:          3.8Gi          0B       3.8Gi
hiba@ubuntu:~$
```

15. lsblk

Explanation: Lists storage devices and their partitions.

Example: `lsblk`

```
hiba@ubuntu:~$ lsblk
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
fd0      2:0    1     4K  0 disk
loop0    7:0    0     4K  1 loop /snap/bare/5
loop1    7:1    0    20M  1 loop /snap/desktop-security-center/151
loop2    7:2    0  66.8M  1 loop /snap/core24/1643
loop3    7:3    0 260.3M  1 loop /snap/firefox/8763
loop4    7:4    0  16.5M  1 loop /snap/firmware-updater/226
loop5    7:5    0  14.2M  1 loop /snap/desktop-security-center/206
loop6    7:6    0 614.5M  1 loop /snap/gnome-46-2404/164
loop7    7:7    0   1.5M  1 loop /snap/hwctl/123
loop8    7:8    0   1.5M  1 loop /snap/hwctl/131
loop9    7:9    0   402M  1 loop /snap/mesa-2404/1839
loop10   7:10   0  91.7M  1 loop /snap/gtk-common-themes/1535
loop11   7:11   0  18.4M  1 loop /snap/prompting-client/228
loop12   7:12   0  18.8M  1 loop /snap/prompting-client/222
loop13   7:13   0  50.1M  1 loop /snap/snapd/27710
loop14   7:14   0  11.8M  1 loop /snap/snap-store/1390
loop15   7:15   0   828K  1 loop /snap/snapd-desktop-integration/391
sda      8:0    0    30G  0 disk
├─sda1   8:1    0     1M  0 part
└─sda2   8:2    0    30G  0 part /
sr0     11:0    1 109.6M  0 rom  /run/media/hiba/CDROM
sr1     11:1    1     6G  0 rom  /run/media/hiba/Ubuntu 26.04.1 LTS amd64
hiba@ubuntu:~$
```

de or press Ctrl+G.

16. /proc

Explanation: A virtual filesystem containing process and kernel information. It is a location, not a command by itself.

Example: `cat /proc/uptime`

```
hiba@ubuntu:~$ cat /proc/uptime
1621.51 1453.63
hiba@ubuntu:~$
```

17. /sys

Explanation: A virtual filesystem exposing device and kernel information. It is a location, not a command by itself.

Example: `ls /sys`

```
hiba@ubuntu:~$ ls /sys
block  bus  class  dev  devices  firmware  fs  hypervisor  kernel  module  power
hiba@ubuntu:~$
```

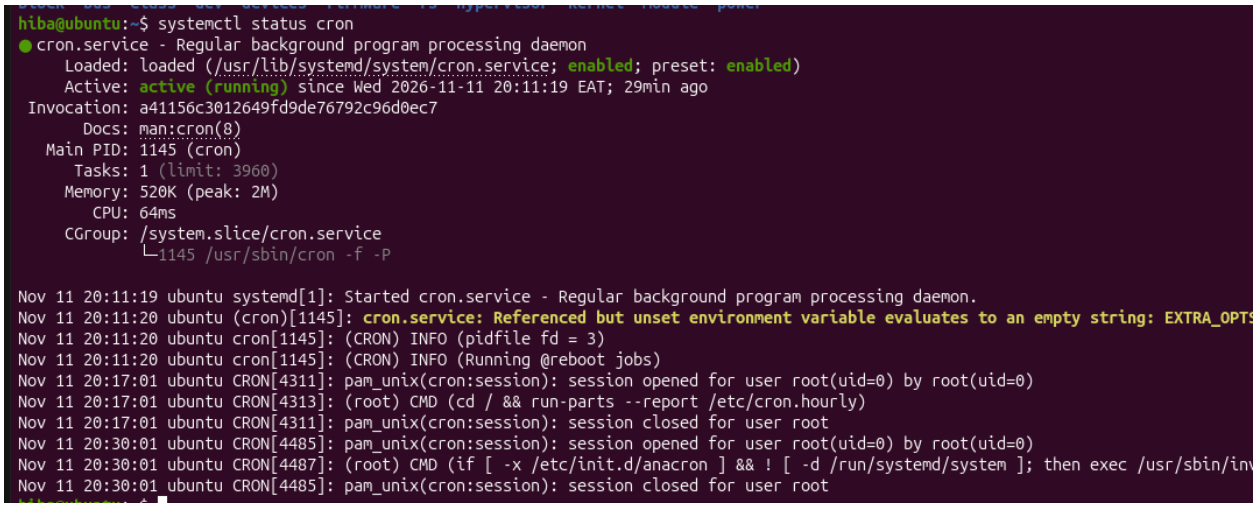
side or press Ctrl+G.

4.4 Services with systemd

18. systemctl status

Explanation: Checks the current status of a system service.

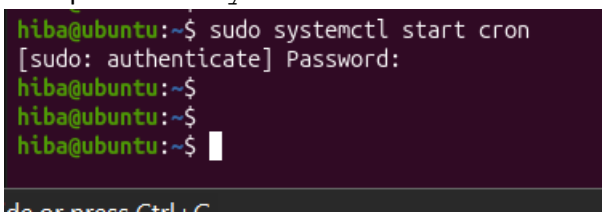
Example: `systemctl status cron`



19. systemctl start

Explanation: Starts a system service.

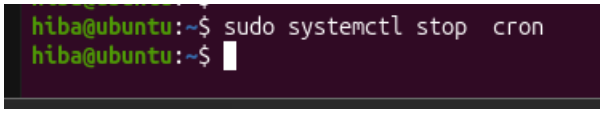
Example: `sudo systemctl start cron`



20. systemctl stop

Explanation: Stops a running system service.

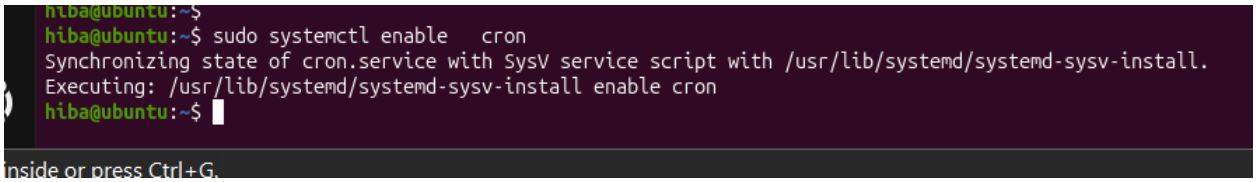
Example: `sudo systemctl stop cron`



21. systemctl enable

Explanation: Configures a service to start automatically at boot.

Example: `sudo systemctl enable cron`

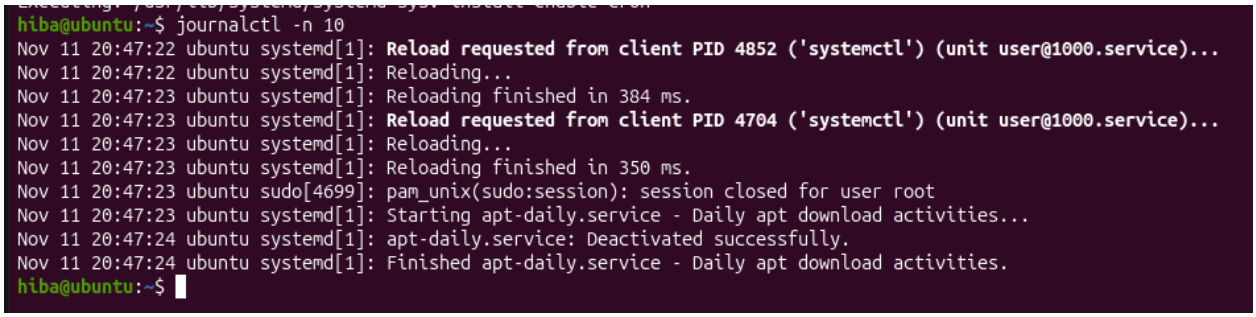


4.5 Logs, Boot and Troubleshooting

22. journalctl

Explanation: Displays logs collected by systemd. The -n option limits the number of lines.

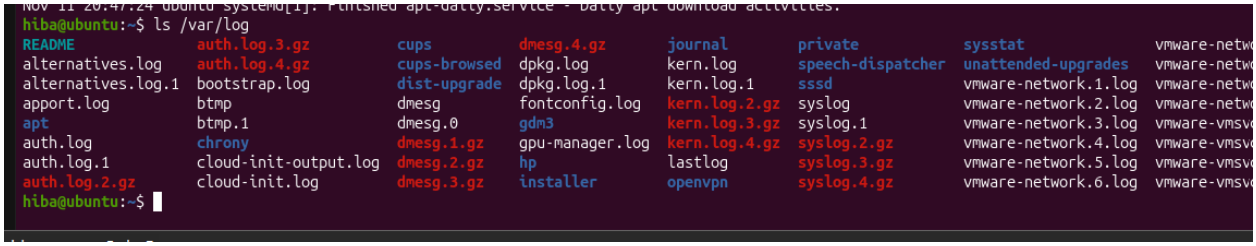
Example: `journalctl -n 10`



23. /var/log

Explanation: A directory where many system and application log files are stored. It is a location, not a command.

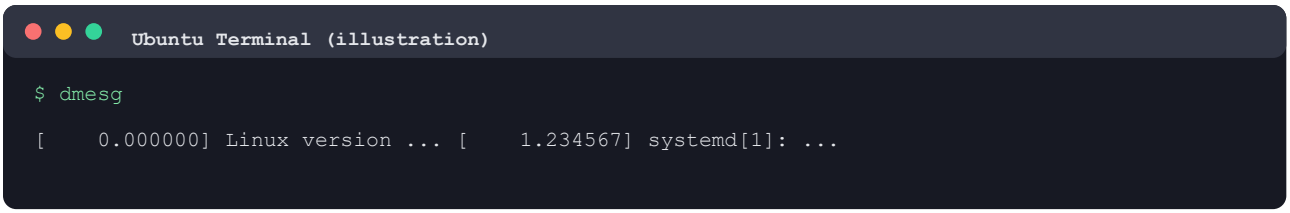
Example: `ls /var/log`



24. dmesg

Explanation: Displays kernel ring-buffer messages, often useful for hardware and boot troubleshooting.

Example: `dmesg`



25. shutdown

Explanation: Shuts down or restarts the computer. Use carefully; this example shuts down immediately.

Example: `sudo shutdown -h now`



Important: htop may not be installed by default. If Ubuntu says “command not found”, install it with `sudo apt install htop` (requires internet and permission). The paths `/proc`, `/sys` and `/var/log` are not standalone commands; examples show commands used to inspect them.